Project Proposal

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	25
Understanding	3					
Analysis	4					
Implementation	8					
Report Writing	10					
Total obtained mark						
Comments						

Semester: FALL 2026

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Adaptive AI Logistics & Operations Intelligence Platform

Project Proposal

1. Introduction

1.1 Background Overview

Organizations such as businesses, educational institutions, NGOs, warehouses, retailers, and distributors regularly purchase, store, consume, transport, and manage resources. A university, for example, may purchase stationery, computers, laboratory equipment, furniture, cleaning supplies, and other resources throughout the year. These activities also involve supplier prices, transportation costs, storage expenses, employee-related costs, budgets, and historical consumption.

In many small and medium-sized organizations, these activities are recorded using spreadsheets, paper-based records, or basic inventory and accounting software. Such systems are useful for storing information, but they generally do not use historical operational data to provide intelligent planning and decision support.

The proposed Adaptive AI Logistics & Operations Intelligence Platform addresses this problem by storing organizational operational data and applying AI to answer three levels of questions: What happened? What is likely to happen? What should we consider doing?

The central research concept is Organizational Memory: the system uses historical transactions together with previous organizational decisions, corrections, feedback, and outcomes to build an organization-specific behavioral profile.

1.2 Problem Statement

- Poor procurement decisions: quantities and purchasing times are often estimated manually, which can lead to overstocking, unnecessary expenditure, or shortages.
- Difficulty predicting demand: organizations may not systematically analyze historical consumption patterns to estimate future requirements.
- Price uncertainty: changes in supplier and market prices may be difficult to identify and interpret manually.
- Late detection of unusual activity: unexpected spending, transactions, or inventory changes may only be noticed after they affect the organization.
- Lack of personalized recommendations: generic forecasting or reorder rules do not consider how a particular organization actually makes decisions.
- Fragmented operational information: purchases, inventory, finance, suppliers, transportation, and consumption may be stored separately.
- Limited access to intelligent tools: advanced supply-chain intelligence systems may be costly or complex for smaller organizations and institutions.

The proposed system is limited to forecasting, recommendation, and analysis based on data provided by the organization. It does not execute purchases or control physical operations.

1.3 Objectives

General Objective: To develop an AI-powered logistics and operations decision-support platform that helps organizations manage resources, analyze operations, forecast future requirements, and make procurement decisions by learning from their own historical data and decisions.
Specific Objectives:

- Develop a centralized platform for organizational operational data.
- Record and analyze purchases, sales/consumption, inventory, suppliers, transportation, expenses, budgets, and related transactions.
- Forecast item-level demand using historical consumption data.
- Analyze historical and current price data to identify price trends.
- Generate procurement recommendations regarding what to purchase, how much to purchase, and when to purchase.
- Detect unusual or anomalous transactions and provide supporting reasons.
- Provide financial analysis such as profit/loss for businesses and budget utilization for institutions.
- Provide a natural-language AI assistant using authorized organizational data.
- Record user decisions, corrections, and feedback on AI recommendations.
- Build an organization-specific memory/profile from historical decisions and outcomes.
- Compare generic, organization-history-based, and personalized AI approaches.
- Evaluate whether organization-specific learning improves forecasting and recommendation performance.

The central research objective is to investigate whether organizational behavioral learning can improve AI-based operational recommendations.

1.4 Scope

In scope:

- Organization registration, setup, and organization type selection.
- Role-based access control and user management.
- Product/resource, inventory, purchase, sales/consumption, and supplier management.
- Supplier price history, transportation and logistics cost recording.
- Expense and budget management.
- Financial summaries and department-level analysis.
- CSV data import/export and market-price data management.
- Demand forecasting, inventory risk analysis, and price trend analysis.
- Procurement recommendations and supplier comparison.
- Anomaly detection and AI recommendation feedback.
- Organizational Memory and personalized recommendations.
- Natural-language AI assistant.

- Dashboards, alerts, reports, audit history, and what-if analysis.

Organization types may include Business, Educational Institution, NGO, Warehouse, Retailer, Distributor, and other configurable organizations.
Out of scope for the initial MVP:

- Fully autonomous purchasing and payment processing.
- Autonomous physical logistics operations.
- Guaranteed future price prediction.
- High-frequency trading or investment decisions.
- Replacement of professional decision-makers.
- Autonomous route execution.
- Large-scale mobile application development before the web system is stable.

All AI outputs are recommendations subject to human review and approval.

1.5 Stakeholders

Stakeholder	**Role**	**Interest in the System**
Administrators / Managers	Review organizational information and recommendations	Better planning and decision support
Procurement Officers	Manage purchasing and supplier information	Appropriate quantity, timing, and supplier decisions
Store / Inventory Officers	Maintain inventory and transaction records	Accurate stock information and reduced shortages
Finance / Accounts Staff	Manage expenses, budgets, and financial information	Clear financial summaries and early warnings
Organization Owners / Institutional Authorities	Review organizational performance	Better resource and budget control
Department Heads	Monitor departmental consumption and expenditure	Department-level visibility and planning
System Administrator	Manage users, roles, configuration, and security	Secure and stable operation
Suppliers	Provide goods and price information	Structured procurement interactions
Course Supervisor / Evaluators	Evaluate the academic project	Technical quality and measurable research contribution

1.6 Proposed Solution

The proposed platform combines an operational management system with an AI intelligence layer.
1. The organization registers and selects its organization type.
2. Authorized users configure departments, locations, resources, suppliers, and organizational information.
3. Users enter or import historical purchases, inventory, consumption/sales, expenses, supplier prices, and related records.
4. The system stores the information in a structured organizational database.

5. The demand forecasting module predicts future demand.
6. The price analysis module identifies historical and current price trends.
7. The procurement module combines forecasts, current stock, supplier information, prices, and other factors to generate recommendations.
8. The user reviews, accepts, modifies, or overrides the recommendation.
9. The final decision is recorded.
10. The personalization layer compares the recommendation with the actual organizational decision.
11. The system updates the organization-specific behavioral profile when sufficient evidence is available.
12. The anomaly detection module identifies transactions that significantly differ from historical patterns.
13. The AI assistant answers natural-language questions using authorized organizational data.

Core learning cycle: Historical Data → AI Prediction → Recommendation → User Decision → Outcome/Feedback → Organizational Memory → Improved Organization-Specific Recommendation

1.7 AI Features

AI Feature	**Main Input**	**Output**	**Manual Work Reduced**
Demand Forecasting	Historical consumption and dates	Future demand forecast	Estimating future quantities
Inventory Risk Analysis	Stock and forecast	Shortage/overstock risk	Manual stock monitoring
Price Trend Analysis	Historical/current prices	Trend and percentage change	Manual price comparison
Procurement Recommendation	Forecast, stock, prices, suppliers	Suggested quantity and timing	Manual purchasing estimation
Supplier Comparison	Supplier prices/history	Comparable supplier information	Manual supplier comparison
Anomaly Detection	Transaction history	Unusual transaction flags	Manual record scanning
Organizational Memory	Recommendations, decisions, feedback, outcomes	Organization-specific adjustment	Repeated manual correction
Financial Intelligence	Revenue, expenses, budgets, inventory	Financial/budget insights	Manual calculation
Data-Grounded Assistant	User question and authorized data	Context-specific answer	Manual searching/report preparation
What-If Analysis	Operational data and assumptions	Scenario outcomes	Manual scenario calculation

Demand forecasting may use time-series and machine-learning techniques such as statistical forecasting, regression, Prophet or equivalent forecasting methods, and gradient-boosting models. Price analysis will examine historical and current prices to identify trends and percentage changes. Procurement recommendations may consider forecasted demand, current inventory, supplier prices, price trends, transportation costs, and previous organizational decisions.

Organizational Memory records recommendations, user decisions, modifications, overrides, feedback, and outcomes. When sufficient history exists, this information can produce an organization-specific adjustment. With insufficient history, the generic baseline is used.

Anomaly detection may use statistical thresholds, Isolation Forest, or other unsupervised methods. An anomaly is a review signal, not proof of fraud or wrongdoing.
The natural-language assistant will use retrieval/function calling over authorized organizational data so that answers are grounded in actual records.

2. System Requirements

2.1 Functional Requirements

ID	Requirement
FR1	The system shall allow users to register an organization and select its organization type.
FR2	The system shall support role-based user accounts and authorization.
FR3	The system shall allow administrators to manage users, departments, locations, and organizational settings.
FR4	The system shall allow users to create and manage products/resources.
FR5	The system shall allow users to record inventory and inventory transactions.
FR6	The system shall allow users to record and edit purchases and purchase items.
FR7	The system shall allow users to record sales or internal consumption.
FR8	The system shall allow users to manage supplier information and supplier prices.
FR9	The system shall allow users to record transportation and logistics information.
FR10	The system shall allow users to record organizational expenses.
FR11	The system shall support budget creation and budget transactions.
FR12	The system shall calculate profit/loss for business organizations.
FR13	The system shall calculate budget utilization and remaining budget for institutions.
FR14	The system shall provide department-level spending and operational summaries.
FR15	The system shall allow historical data to be imported/exported using supported formats such as CSV.
FR16	The system shall allow market-price data to be stored and analyzed.
FR17	The system shall provide dashboards containing operational, financial, inventory, and AI information.

FR18	The system shall provide alerts and notifications for relevant events.
FR19	The system shall maintain an audit history of important system activities.
FR20	The system shall provide what-if analysis for supported scenarios.

2.2 AI Requirements

ID	Requirement
AI1	The system shall forecast future demand for supported items.
AI2	The system shall provide forecast uncertainty/error information where supported.
AI3	The system shall analyze historical and current price trends.
AI4	The system shall recommend procurement quantities based on available operational information.
AI5	The system shall recommend appropriate procurement timing or planning periods.
AI6	The system shall record the user's final decision for an AI recommendation.
AI7	The system shall record feedback and outcomes associated with recommendations.
AI8	The system shall maintain an organization-specific behavioral profile based on sufficient evidence.
AI9	The system shall use a generic model when insufficient organization-specific history is available.
AI10	The system shall identify transactions that significantly deviate from historical patterns.
AI11	The system shall provide supporting reasons for anomaly flags where possible.
AI12	The AI assistant shall answer questions using only data the requesting user is authorized to access.
AI13	The AI assistant shall distinguish historical facts from forecasts, assumptions, and recommendations.
AI14	The system shall allow users to override AI recommendations.
AI15	The system shall support evaluation of generic, organization-specific, and personalized approaches.

2.3 Non-Functional Requirements

ID	Requirement

NFR1	The system shall restrict organizational data to authorized users.
NFR2	The system shall maintain logical data isolation between organizations.
NFR3	Dashboard pages should provide acceptable response times for supported datasets.
NFR4	AI assistant responses should be returned within a reasonable period under normal conditions.
NFR5	The system shall be usable through modern web browsers.
NFR6	The system shall provide a responsive interface.
NFR7	The system shall maintain data integrity during transactions and updates.
NFR8	Important user and system actions shall be auditable.
NFR9	AI outputs shall provide appropriate explanations or supporting factors where possible.
NFR10	The system shall support backup and recovery mechanisms.
NFR11	The system shall be maintainable through modular architecture.
NFR12	The system shall support unit and integrated testing.
NFR13	Sensitive organizational information shall be handled using appropriate privacy and security practices.
NFR14	The system shall support localization requirements relevant to the target environment, including BDT where applicable.

3. Tools and Technologies

Category	**Proposed Technology**
Frontend	React / JavaScript or TypeScript
Backend	Django + Django REST Framework or equivalent REST backend
Database	PostgreSQL
AI/ML	Python
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn
Forecasting	Prophet, Statsmodels, or equivalent models
Advanced ML	XGBoost / LightGBM where appropriate
LLM	LLM API with function calling / retrieval
API Testing	Postman
UI/UX Design	Figma
Version Control	Git / GitHub
Development	Jupyter Notebook and standard IDE
Deployment	Containerized deployment where appropriate

4. Project Timeline and Work Plan

Milestone	Weeks	Main Deliverable
M1: Planning and Requirements	1–2	Final requirements, architecture, database design, research plan
M2: UI/UX and System Foundation	2–4	Authentication, organization setup, roles, UI foundation
M3: Core Operations	4–6	Products, inventory, purchases, consumption, suppliers
M4: Finance and Logistics	6–8	Expenses, budgets, financial calculations, transportation/logistics
M5: Data and Analytics	8–9	Dashboards, reports, historical analysis, market-price data
M6: AI Baseline	9–10	Demand forecasting, price analysis, baseline models
M7: AI Recommendations	10–11	Procurement recommendation engine
M8: Organizational Memory	11–12	Feedback recording, behavioral profile, personalization
M9: Anomaly Detection and AI Assistant	12–13	Anomaly module and data-grounded assistant
M10: Evaluation and Testing	13–15	Model comparison and system testing
M11: Final Delivery	15–16	Documentation, final report, demonstration, presentation

4.1 Task Division

Member	Responsibilities
Member 1	Backend, database, authentication, organization/RBAC, inventory, procurement, finance, system integration
Member 2	Machine learning, demand forecasting, price analysis, recommendation engine, Organizational Memory, model evaluation
Member 3	Frontend/dashboard, anomaly detection, AI assistant, testing, visualization, documentation

5. Optional Sections

5.1 Existing System

Existing approaches generally include spreadsheets and manual records, basic inventory/accounting systems, and enterprise ERP/supply-chain platforms. The first are flexible but manual; the second commonly rely on fixed rules; and the third can be costly and complex. The proposed platform combines

operational management with forecasting, recommendations, anomaly detection, and organization-specific learning.

5.2 System Architecture

The proposed system follows a layered architecture: Web/Mobile-Ready Interface → REST API/Application Backend → Operational and Business Logic → AI/ML Intelligence Layer → PostgreSQL Organizational Database.
The AI layer contains separate components for Demand Forecasting, Price Analysis, Inventory Risk, Procurement Recommendation, Anomaly Detection, Organizational Memory, Financial Analysis, and the Natural-Language Assistant.
Core feedback loop: Historical Transactions → AI Forecast/Recommendation → User Decision → Feedback/Outcome → Organizational Memory → Organization-Specific Adjustment → Future Recommendation.

5.3 Evaluation Plan

Three approaches will be compared: (1) a generic baseline, (2) a model using organization history, and (3) a personalized model that additionally learns from user decisions, feedback, and outcomes.
Forecast performance may be measured using MAE, RMSE, and MAPE. Where suitable labeled or simulated data is available, anomaly detection may be evaluated using precision, recall, F1-score, and ROC-AUC. Recommendation quality may be evaluated by comparing recommendations with actual organizational decisions and by testing controlled simulated organizational behaviors.
The evaluation will report measured results and will not assume that personalization improves performance without evidence.

5.4 Ethical, Legal, and Social Considerations

- Data privacy: organizational data must remain isolated and accessible only to authorized users.
- Human oversight: AI outputs are recommendations and the user remains responsible for consequential decisions.
- Explainability: important recommendations and anomaly flags should provide supporting factors where possible.
- Bias and organizational habits: personalization may reproduce inefficient behaviors such as chronic under-stocking.
- AI reliability: the assistant should retrieve actual organizational records and state when information is insufficient.
- Data sources: user-provided, public, and controlled simulated datasets may be used.

5.5 Limitations

- Forecast quality depends on the quantity and quality of historical data.
- New organizations may initially have insufficient history for personalization.

- Price analysis cannot guarantee future market prices.
- Public and simulated datasets may not fully represent real organizational behavior.
- Organizational Memory may reproduce inefficient decision habits.
- AI recommendations require human validation.
- Assistant performance depends on the quality and structure of organizational data.

5.6 Future Work

- Automated purchase-order generation.
- Advanced supplier evaluation and ranking.
- Multi-branch organizational management.
- Mobile application.
- Automated market-price data collection.
- Real-time vehicle/GPS tracking.
- Route and delivery optimization.
- Automated alerts.
- Advanced what-if simulation.
- ERP/accounting integration.
- More advanced organizational behavior modeling.
- More sophisticated feedback-based recommendation methods.

5.7 Final Project Definition

The Adaptive AI Logistics & Operations Intelligence Platform is not intended to be only an inventory-management system, accounting application, or chatbot. It consists of three connected layers.

1. Operational Foundation: purchases, inventory, consumption/sales, suppliers, prices, expenses, budgets, transportation, deliveries, and historical transactions.

2. Intelligence Layer: forecasts, price trends, inventory-risk insights, procurement recommendations, financial analysis, anomaly alerts, and natural-language answers.

3. Organizational Learning Layer: AI recommendations, user decisions, corrections, feedback, and actual outcomes are recorded to construct an organization-specific behavioral profile.

Central Research Question: Does incorporating an organization's historical decisions and feedback into an AI-based logistics and procurement system improve its recommendations compared with a generic model?

The intended system follows the cycle: Record → Analyze → Predict → Recommend → Observe Decision → Learn → Personalize.

The AI remains a decision-support mechanism, while humans remain responsible for consequential organizational decisions.